CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

Fifth Semester of B. Tech. Examination (CE & IT), Dec 2011

CE301: Parallel Computing

Date: 13.12.11, Tuesday Time: 10:00a.m. To 01:00p.m. Maximum Marks: 70

Instructions:

(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question Number.
(3) Figures to right indicate marks.
(4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions. [11]

[A] The PRAM model which allows either concurrent read or concurrent write at the same time is called ______________ [1]

[B] What is the difference between Parallel Computing and Pipelining? [1]

[C] Given a task that can be divided into **m** subtask, each requiring **one** unit of time, How much time is required for an **m-staged** pipeline to process **n** tasks? [1]

[D] What are the factors affecting the ideal speedup? [2]

[E] How to calculate Time complexity of an algorithm for parallel computers? [2]

[F] What are the attributes of thread? [2]

[G] What is contention? How it can be removed? [2]

Q-2 Answer the following questions. [12]

[A] Differentiate between data parallelism and temporal parallelism. [6]

[B] Explain requirements of multiprocessor operating system. [6]

OR

[A] Differentiate: Shared memory Vs. Dynamic memory programming model. [4]

[B] Explain Flynn's classification of parallel computers. [4]

[C] What is thread? Compare it with process. Explain attributes of thread. [4]

Q-3 Answer the following questions.(Attempt any Three) [12]

[A] Explain a spawning of children and group communication for message passing model?

[B] Write the architectural differences between UMA and NUMA Architecture.

[C] Derive speed up equation for data parallelism.

[D] What is condition variable? Write the advantages of it. Give example using thread.

SECTION-II

Q-4 Answer the following questions. **[11]**

[A] State true or false: More CPU cycle is wasted in creating a thread than a child process through fork system call. [1]

[B] Define Zombie Process. [1]

[C] List the header files required to be included for using shmget() function. [1]

[D] What is the difference between wait() and waitpid()? [1]

[E] How many total numbers of processes (Including Parent) will be created from following program? Justify your answer by taking suitable diagram. [1]

```
main()
{
 for(i=0;i<3;i++)
   for(i=0;j;2;j++)
      fork();
}
```

[F] What are the criteria to for selecting either loop splitting or self-scheduling method of parallelism? [2]

[G] Brief induction variable with example. [2]

[H] List the steps required to develop Server Application in DCE. [2]

Q-5 Answer the following questions. **[12]**

[A] Write the implementations of following functions:- [6]
I. barrier_init() II. barrier()

[B] Write a thread program for addition of two 4 X 4 matrices and store result into third matrix [6]

OR

[A] Explain in detail: PVM Architecture [6]

[B] Write a parallel program in 'C' to create histogram from an array using self-scheduling. [6]

Q-6 Answer the following questions.(Attempt any Three) **[12]**

[A] Write a parallel program in C to compute n! using loop splitting.

[B] What is a race condition? How can it be avoided?

[C] Explain Forward and backward dependency with example. How it can be solved?

[D] Write a parallel program in C to find the number is prime or not.